

Week 7 Weekly Report

Week 7 extended the original selected models with counterfactual stress tests, VIX-focused interpretation, a complete market-feature updater, and a Streamlit prototype. VIX remains a market volatility indicator and is not relabeled as news sentiment.

1 Work Completed

- Measured VIX-related SHAP contribution for the selected Random Forest and Linear Regression models.
- Applied the required 50% volatility spike and absolute 2 percentage-point rate hike to the structural BSM model.
- Applied the same counterfactual inputs to both machine-learning pricing routes.
- Recorded the absence of a licensed or reproducible news-sentiment series instead of fabricating a feature.
- Built a JPM, Cboe VIX, FRED DGS10 and DGS1 updater with bounded retries, independent source caches, and complete feature reconstruction at a common completed date.
- Built and locally health-tested a Streamlit pricing and sensitivity prototype.

2 Deliverables

- Week7_Advanced_Analysis.ipynb
- sensitivity_analysis.py
- data_update.py
- Streamlit app.py
- tool_config.json
- Comprehensive Sensitivity Analysis Report
 - stress, SHAP, and data-status outputs

3 Progress Summary

The required structural shocks, ML counterfactuals and sensitivity report are complete. Both applications refresh on opening, every 15 minutes while active, and on manual request. Tests verified complete original-feature reconstruction and explicit source-failure fallback. The original trained models and historical results remain unchanged.

4 Experimental Results

Scenario	BSM	Approach 1	Approach 2
Base	99.2617	91.7072	0.0000
50% volatility spike	115.6683	92.0449	0.0000
2% rate hike	101.3317	94.5557	0.0000
Combined shock	116.9177	94.8010	0.0000

Figure 1 Structural chooser stress scenarios



5 Issues Encountered

- Public endpoints can time out or publish on different dates. Each request uses a 12-second timeout and at most two attempts. Per-source caches preserve available histories, and incomplete updates retain a labeled complete snapshot. FRED credentials stay outside logs and saved data.
- A news-sentiment feature requires a defined source, timestamp policy, and leakage controls. These inputs were not present in Weeks 1 and 2, so sentiment remains a documented data gap.
- ML counterfactuals can combine inputs that are uncommon in the training sample; results must be read with the training-domain limits.

6 Plan for Week 8

- Package BSM and the best ML model into a self-contained final tool.
- Display an empirical proxy-error margin and model limitations.
- Prepare the deployment checklist, final report, presentation, and demo script.